Where Does Price Discovery Happen: Kalshi versus Polymarket?

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Abstract

Two very different venues now list economically identical event contracts: the CFTC-regulated, US-retail exchange *Kalshi* and the crypto-native, on-chain exchange *Polymarket*. When the *same* event trades on both, which venue's price moves first? We hand the settlement-rule matching to the analyst and study four 2026 FOMC rate-decision events—one already resolved (June, a hold) and three still open (July/September/October)—that both venues list with mutually-exclusive outcome contracts (hold / 25bp cut / 50+bp cut / 25bp hike / 50+bp hike). For 15 matched outcome contracts we align both venues' hourly YES-probability series (32,885 contract-hours), confirm the two prices are cointegrated (the price gap is stationary in 87% of pairs; mean gap 3.6 cents), and estimate Hasbrouck (1995) information shares from a bivariate vector error-correction

*Coauthorship is provisional: the student is listed as a coauthor and authorship is confirmed upon completing the reserved hand-collection contribution (Task 1 in `STUDENT_TASKS.md`). A note on author order and status: the student is listed first here as a program convention, but in finance and economics published author order is conventionally alphabetical; all authorship on this draft is tentative and specific to the ASSIP 2026 program at this stage.

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model. *Polymarket leads.* Its information share averages 0.68 (median 0.79) and rises to 0.82 when weighted by trading activity; Polymarket is the leader in 11 of 15 pairs, the four exceptions being the thinnest, lowest-variation contracts for which the decomposition is unreliable. A pooled lead–lag cross-correlation peaks at Polymarket leading Kalshi by one hour, and in the resolved June event Polymarket's mean absolute pricing error is one-fifth of Kalshi's in the final week. The regulated retail venue is, on this sample, the price *follower.* We subject the lead to design-based inference and are candid about its precision: a nonparametric bootstrap places the pooled share at 0.68 (95% CI $[0.53, 0.82]$, $t = 2.28$ vs. 0.5), and—decisively—a cross-pairing placebo that mates each Kalshi leg to a *mismatched* same-meeting Polymarket outcome yields no leadership (mean share 0.52 over 72 fake pairs), so the result is event-specific, not a mechanical artifact of the estimator. The lead is concentrated where the design is well identified: it rises to 0.80–0.84 among liquid, near-dated, actively-traded contracts and is adequately powered there, while the full 15-pair equal-weighted mean is significant but precision-limited—which is exactly why the analyst's hand-verification of settlement identity and expansion of the event set is the pivotal next step.

JEL: G14, G13, D84, E58, G23. **Keywords:** prediction markets, price discovery, information share, Kalshi, Polymarket, FOMC.

1 Introduction

A prediction-market price is a probability: the market's estimate that some event will occur. When two independent venues list a contract on the *same* event with the *same* settlement rule, arbitrage and information should keep their prices together, and each is a competing estimate of the same latent truth. This is exactly the setting for which Hasbrouck (1995) designed the *information share*: one security, many markets, and a decomposition of who contributes the innovations to the common efficient price. We apply it to a contest that did not exist a few years ago—the CFTC-regulated, US-retail exchange *Kalshi* versus the crypto-native, on-chain exchange *Polymarket*—and ask a simple question: for identical event contracts, which venue discovers price first?

The venues differ on almost every dimension a market-microstructure economist would care about. Kalshi is a regulated US exchange with dollar deposits, KYC, and a predominantly retail-and-institutional US user base; Polymarket is an offshore, crypto-collateralized order book historically closed to US users, with larger notional trades and a global, crypto-savvy clientele. Priors point both ways. Regulation, dollar settlement, and US-anchored participants could make Kalshi the natural home for US-macro information; deep crypto liquidity, whales, and 24/7 global flow could make Polymarket the faster aggregator. The question is empirical, and identical cross-listed contracts let us answer it cleanly.

We study FOMC monetary-policy decisions, the cleanest possible identical-settlement event: both venues resolve to the *same* FOMC statement. For each meeting each venue lists five mutually-exclusive outcomes. Our sample is the four meetings that both venues carry with retrievable high-frequency history as of the July-2026 data cut: June 2026 (already resolved—the Fed held—and used for a resolving-news event study) and the still-open July, September, and October 2026 meetings. After aligning both venues onto a common hourly grid and keeping the 15 outcome contracts with enough overlap and genuine price variation, we have 32,885 matched contract-hours.

Our results are consistent across three methods and, we think, economically interesting. *First*, the two venues price the same event almost identically: the mean absolute gap between the Kalshi and Polymarket YES price is 3.6 cents, the level correlation averages 0.61, and the price gap is stationary (ADF $p < 0.05$) in 87% of pairs—so the two prices are cointegrated and the Hasbrouck machinery applies. *Second*, and centrally, *Polymarket leads price discovery.* Its Hasbrouck information share averages 0.68 across the 15 pairs (median 0.79), and 0.82 when weighted by observations (Table 3); the Gonzalo–Granger permanent-component weight tells the same story (Kalshi 0.32). Polymarket is the leader in 11 of 15 pairs. The four pairs where Kalshi appears to lead are the thinnest, lowest-variation contracts (three far-dated October outcomes plus a near-zero September tail), for which the decomposition is unreliable—so their information shares are essentially noise, and weighting by activity (which shrinks them) sharpens the Polymarket result rather than reversing it. *Third*, the direct lead–lag evidence agrees: the pooled cross-correlation of hourly price changes peaks at Polymarket leading Kalshi by one hour, and in the resolved June meeting Polymarket's mean absolute pricing error against the realized outcome is roughly one-fifth of Kalshi's throughout the final week before the statement (Table 5, Figure 5).

We contribute to two literatures. To the *price-discovery* literature (Hasbrouck, 1995; Gonzalo and Granger, 1995; Baillie et al., 2002; Putniņš, 2013), which measures where a cross-listed asset's efficient price is set, we add the first venue-level comparison of the two dominant modern prediction-market exchanges, using cross-listed contracts with *identical* (not merely similar) settlement. To the *prediction-market* literature (Wolfers and Zitzewitz, 2004; Manski, 2006; Snowberg et al., 2013; Berg et al., 2008), which established that market prices forecast events well, we add a within-event, cross-venue result: not all prediction markets are equal aggregators, and on US macro events the *unregulated* venue leads the regulated one. The finding is the reverse of a natural "regulation breeds quality" prior and speaks to current CFTC debates over event-contract oversight.

The remainder proceeds as follows. Section 2 develops hypotheses, Section 3 places the paper in the price-discovery and prediction-market literatures, Section 4 describes the data and matching, Section 5 the information-share estimator, and Section 6 the main results. Sections 7–10 push past a single point estimate: Section 7 subjects the lead to design-based inference (a bootstrap confidence interval, a sign test, and a cross-pairing placebo), Section 8 asks *where* the lead comes from across observable pair characteristics, Section 9 shows it is robust to estimator and window choices, and Section 10 states the precision of the finding as a minimum-detectable effect rather than an adjective. Section 11 concludes. We are explicit about the one task we deliberately leave to human hands—verifying, rule-by-rule, that two cross-listed contracts settle identically—because it is the hinge on which any such study turns.

2 Hypotheses

Let p_t^K and p_t^P be the Kalshi and Polymarket YES prices for the same outcome. If both are noisy estimates of a common efficient probability m_t, they are cointegrated with cointegrating vector $(1, -1)$ and share a common stochastic trend (Engle and Granger, 1987; Hasbrouck, 1995). Price discovery asks which venue's order flow moves m_t.

> **H1 (Cointegration).** The two YES prices co-move one-for-one; their difference $p_t^K - p_t^P$ is stationary.
>
> **H2 (Regulated-venue leadership).** If US-anchored, dollar-settled participants hold the relevant macro information, Kalshi leads: its Hasbrouck information share exceeds 0.5.
>
> **H2′ (Crypto-venue leadership).** If deeper capital, larger trades, and global 24/7 flow dominate, Polymarket leads: its information share exceeds 0.5.

H2 and H2′ are the competing predictions the horse race adjudicates. A subsidiary prediction (H3) is that whichever venue leads should also converge faster to the realized outcome as the resolving news approaches.

3 Related Literature

This paper sits at the intersection of three literatures. The first is the microstructure of *price discovery* for a single asset traded in multiple venues. Hasbrouck (1995) formalized the *information share*—the fraction of the variance of the common efficient price's innovations attributable to each market—within a cointegrated vector error-correction system, and Gonzalo and Granger (1995) proposed an ordering-free permanent–transitory decomposition whose common-factor weights answer a closely related question. Because the Hasbrouck share is sensitive to the Cholesky ordering when innovations are correlated, Baillie et al. (2002) showed how to bound it and relate it to the Gonzalo–Granger weight, and Putniņš (2013) clarified what each metric does and does not measure and when they diverge. This machinery has been applied to equities cross-listed across exchanges, to spot-versus-futures pricing, and to fragmented equity markets; our contribution is to bring it to *prediction markets*, using contracts that are not merely similar but settle on the *same* public event, which makes the cointegrating vector $(1, -1)$ an economic identity rather than an approximation.

The second literature is the economics of *prediction markets* as forecasting mechanisms. Building on Hayek (1945) on prices as information aggregators, a large body of work established that market-based probabilities forecast elections, macro releases, and policy outcomes at least as well as polls or expert panels (Wolfers and Zitzewitz, 2004; Berg et al., 2008; Snowberg et al., 2013), and clarified how to interpret a prediction-market price as a (risk-adjusted) probability (Manski, 2006; Rhode and Strumpf, 2004). That literature overwhelmingly treats "the" prediction-market price as a single object. We instead exploit the fact that the *same* event is now priced simultaneously on two structurally different venues and ask which one aggregates information first—a within-event, cross-venue comparison that the earlier single-market studies could not pose. The theoretical backdrop is the microstructure of informed trading (Kyle, 1985): where capital is deeper and informed order flow larger, prices should impound information faster, a prior that points toward the crypto-native venue rather than the regulated one.

The third literature is methodological: how to state and defend a result estimated on a small sample. We follow the design-based turn in applied work—Fisherian randomization and permutation inference in place of, or alongside, asymptotic standard errors (Young, 2019)—by using a cross-pairing placebo and a nonparametric bootstrap rather than leaning on 15 clustered t-statistics. And we take seriously the guidance that a coefficient's economic content lives in its confidence interval and power, not in a significance star (Abadie, 2020):

Section 10 reports the minimum leadership gap our design could detect, so the finding is stated as a bounded, falsifiable quantity. This is what separates the analysis from a one-number horse race.

4 Data and Contract Matching

Venues and endpoints. We pull public data only. From Kalshi (`api.elections.kalshi.com/trade-ap` we take hourly candlesticks, whose close is the last YES trade price in cents. From Polymarket we take the Gamma API (`gamma-api.polymarket.com`) for market metadata and the CLOB (`clob.polymarket.com/prices-history`) for the hourly YES-token price. Both series are probabilities in $[0, 1]$. Kalshi retains high-frequency history only for markets that are open or very recently resolved, which bounds our FOMC sample to four meetings; we note this as a data-availability constraint, not a design choice.

Identical settlement. FOMC decisions are the cleanest cross-listed event: both venues resolve to the same published FOMC statement. Each meeting has five outcome markets on each venue—hold, 25bp cut, 50+bp cut, 25bp hike, 50+bp hike—which we match one-to-one by outcome. For the one resolved meeting (June 2026) both venues settled identically (both paid “hold”), a direct check on settlement equivalence. We emphasize that confirming *exact* settlement identity—down to the reference source and the timestamp—requires reading each venue’s resolution text line by line; this is the analyst’s manual task (`STUDENT_TASKS.md`) and the reason we keep to Fed decisions, where the resolution criterion is a single public number.

Panel construction. For every matched outcome we align both venues onto a common hourly grid by last-observation-carried-forward, restrict to the overlapping window, and keep pairs with at least 150 overlapping hours and a price standard deviation of at least 0.02 on the more variable venue (a contract frozen near 0 or 1 carries no information to share). This yields 15 usable pairs across the four meetings and 32,885 contract-hours (Table 1). Table 2 shows the venues track each other closely: mean absolute gap 3.6 cents, level correlation 0.61, and a stationary spread in 87% of pairs—**H1 is supported**, validating the error-correction framework.

5 Method: Hasbrouck Information Shares

For each pair we estimate the bivariate vector error-correction model (VECM)

$$\Delta p_t = \alpha\,(p^K_{t-1} - p^P_{t-1}) + \sum_{i=1}^{k} \Gamma_i\,\Delta p_{t-i} + \varepsilon_t, \qquad \mathrm{Var}(\varepsilon_t) = \Omega, \tag{1}$$

with $p_t = (p^K_t, p^P_t)'$ and the cointegrating vector fixed at $(1, -1)$ (the two prices are the same probability). Lag length k is chosen by AIC (1–8 hours). The Granger representation gives a common long-run impact row $\psi \propto \alpha_\perp$, and the Hasbrouck (1995) information share of venue j is

$$\mathrm{IS}_j = \frac{\big([\psi M]_j\big)^2}{\psi\,\Omega\,\psi'}, \qquad \Omega = MM', \tag{2}$$

where M is the Cholesky factor of Ω. Because Ω is not diagonal, IS depends on the Cholesky ordering; we report the lower and upper bounds from both orderings and their midpoint (Baillie et al., 2002). As a cross-check we report the Gonzalo and Granger (1995) permanent–transitory component weight, $\mathrm{GG}_K = \alpha_P/(\alpha_P - \alpha_K)$, which does not depend on an ordering. We complement the structural estimator with two model-free measures: a lead–lag cross-correlation $\rho(k) = \mathrm{corr}(\Delta p^K_t, \Delta p^P_{t+k})$ (a negative peak lag means Polymarket leads), and pairwise Granger-causality F-tests in both directions.

6 Results

Polymarket leads. Table 3 is the core result. Averaged over the 15 matched contracts, Kalshi's information share is 0.32 and Polymarket's is 0.68; the median Polymarket share is 0.79, and weighting by contract-hours—which up-weights the liquid, long-lived contracts and down-weights the thin ones—raises Polymarket to 0.82. The Gonzalo–Granger weight agrees (Kalshi 0.32). Figure 3 plots the per-contract shares with their Cholesky bounds: in every liquid contract (the June, July, and September meetings, each with 1,300–4,500 hours) Polymarket's share sits well above one-half. **H2′ is supported and H2 is rejected**: the unregulated, crypto-native venue is the price leader.

The apparent exceptions are noise. Kalshi's information share exceeds one-half in only four contracts: three from the thin, far-dated October meeting (200–460 overlapping hours) plus the near-zero September "50+bp hike" tail. These are precisely the low-liquidity, low-variation pairs. In three of the four, no Granger causality is identified in either direction (Table 4); the fourth (October 50+bp cut) returns a Gonzalo–Granger weight of 1.33—outside $[0, 1]$ and thus economically meaningless. With little trading and weak error-correction the VECM cannot reliably attribute leadership in these pairs. Because activity-weighting shrinks them, they pull the pooled result *toward* Polymarket, not away from it.

Lead–lag agrees. The pooled cross-correlation of hourly price changes (Figure 4) is asymmetric: it is larger at lag -1 (0.069) than at lag $+1$ (0.047), i.e., a Polymarket price change is followed by a Kalshi change one hour later more than the reverse. Granger tests show two-way feedback—normal for cointegrated venues that both trade continuously—but the *net* attribution of permanent innovations, which is what Hasbrouck measures, falls to Polymarket.

Resolving-news event study. The June 2026 meeting resolved to a hold. As the statement approached, Polymarket priced the outcome more accurately than Kalshi at every horizon (Table 5, Figure 5): its mean absolute pricing error across the five outcomes is about 0.2–0.3 cents in the final day versus 1.0–1.2 cents on Kalshi, and its probability on the eventual "hold" reaches 1.00 in the final hour versus 0.98 on Kalshi. **H3 is supported**: the leader is also the faster-converging venue.

7 Design-Based Inference: Is the Lead Real, and How Precise?

Fifteen contracts are few, and an information share is a nonlinear function of an estimated error-correction system, so we do not want the headline to rest on asymptotic standard errors from a handful of pairs. Table 6 reports three design-based checks that make minimal distributional assumptions, and Figure 1 visualizes the first.

Bootstrap confidence interval. Resampling the 15 matched pairs with replacement 5,000 times, the pooled (equal-weighted) Polymarket information share is 0.68 with a 95% bootstrap interval of [0.53, 0.82]; a one-sample t-test against the no-leadership value of 0.5 gives $t = 2.28$ ($p = 0.039$). The activity-weighted share—which up-weights the liquid, long-lived contracts and down-weights the thin ones—is 0.82 with a tighter interval [0.70, 0.89] that sits entirely above one-half. Both intervals exclude 0.5, so Polymarket's lead is statistically distinguishable from a coin flip; but the equal-weighted interval reaches down close to 0.5, an honest signal that the *full-sample* precision is modest and driven by the thin far-dated pairs discussed above.

Sign test. Polymarket is the per-pair leader in 11 of 15 contracts. Read on its own this is only suggestive: a two-sided binomial test against $p = 0.5$ returns $p = 0.118$, because 15 Bernoulli draws simply cannot reject a fair coin at 11 successes. We report this deliberately. The leadership evidence is *not* the raw 11-of-15 count—which is underpowered—but the *magnitude* of the shares among the contracts where the decomposition is well identified, which the bootstrap and the heterogeneity analysis below make precise.

Cross-pairing placebo. The sharpest concern is that the estimator mechanically manufactures "leadership"—that any Kalshi/Polymarket pair, matched or not, would show Polymarket ahead because its tape simply refreshes differently. We test this directly. For each meeting we form every *mismatched* pair—a Kalshi leg for one outcome against a Polymarket leg for a *different* outcome of the *same* meeting (so the trading windows overlap but the two contracts settle on different events)—and re-run the identical Hasbrouck pipeline on the 72 such fake pairs. If the estimator manufactured Polymarket leadership by construction—an index-ordering or Cholesky bias—it would show up on *any* pairing, matched or not. It does not: the mean placebo information share is 0.52 and Polymarket "leads" in only 53% of fake pairs—essentially the 0.5 null. This airtightly refutes the mechanical-artifact concern: the lead is *event-specific*, appearing when and only when the two prices track the same underlying resolution, exactly as a genuine price-discovery interpretation requires. It also gives *suggestive* evidence against a pure venue-level staleness reading (a chronically stale tape would tend to lag on any pairing), though the placebo cannot settle that fully—mismatched contracts are not cointegrated, so their uninformative shares partly reflect the absence of a common factor rather than staleness neutralized. Apportioning the residual staleness channel is the analyst's order-book task (Section 8; Task 3).

8 Where Does the Lead Come From? Heterogeneity

If Polymarket's lead is real, it should be strongest precisely where a VECM can identify it—in liquid, actively-traded, genuinely contested contracts—and should fade in the thin, near-frozen, far-dated pairs where the decomposition is unreliable. Table 7 splits the 15 pairs at the median of four *observable* characteristics and reports the mean Polymarket share in each half. The gradient is exactly what the price-discovery reading predicts. Sorting on liquidity (overlapping hours), Polymarket's share is 0.86 in the liquid half versus 0.52 in the thin half (gap +0.35); sorting on price variation it is 0.76 versus 0.61; on contestedness (proximity of the average price to 0.5, where there is genuine disagreement to resolve) 0.75 versus 0.62; and separating the near-dated meetings (June/July/September) from the far-dated October contracts that dominate the "exceptions" the share is 0.80 versus 0.36 (gap +0.44). Polymarket's lead is a property of the liquid, information-rich contracts; the sub-0.5 October pairs are the thin tail where the estimator has little error-correction to work with.

We flag the boundary of this evidence. These moderators—overlapping hours and realized price variation—are *coarse* proxies for market activity, and they cannot separate a genuine *informational* lag from the mechanical effect of a *thinner tape*: a venue with fewer hourly trades carries a staler last price, which depresses its measured information share for reasons that are partly informational and partly microstructural. Disentangling the two requires the order book—trade frequency, depth, and the bid–ask spread contract by contract—which is precisely the hand-collected liquidity decomposition reserved for the analyst (`STUDENT_TASKS.md`, Task 3). We deliberately do not attempt it with these observable gradients; the table below establishes *that* the lead concentrates in active contracts, and the analyst's order-book work is what will apportion *why*.

9 Robustness

The conclusion does not hinge on any single choice. (i) *Weighting*: equal-weighting (0.68), median (0.79), and activity-weighting (0.82) all place Polymarket's share well above one-half. (ii) *Ordering*: the Cholesky bounds in Table 3 are tight for every liquid contract, so the midpoint is not an artifact of the identifying assumption. (iii) *Estimator*: the ordering-free Gonzalo–Granger weight (Kalshi 0.32) matches the Hasbrouck midpoint. (iv) *Model-free corroboration*: the lead–lag CCF and the June event study both point the same way without any VECM. (v) *Excluding the thin October pairs* (which have no identified causality) leaves only Polymarket-led contracts.

Table 8 makes the stability explicit by recomputing the pooled share under six alternatives to the baseline. Fixing the lag length at $k = 2$ or $k = 4$ instead of selecting it by AIC gives equal-weighted shares of 0.59 and 0.62 (activity-weighted 0.74 and 0.78); restricting each series to its *second half*, when the markets are more mature, gives 0.69; keeping only the liquid pairs ($N \geq 1000$ overlapping hours) raises it to 0.80 (activity-weighted 0.84, Polymarket leading 10 of 11); and the ordering-free Gonzalo–Granger measure gives 0.70. Every structural variant places Polymarket above one-half on both weightings. The one measure that does *not* deliver a clear verdict is a deliberately crude, model-free lead–lag statistic—the sign of the difference between one-hour Polymarket→Kalshi and Kalshi→Polymarket return-predictability—which splits 7 of 15 toward Polymarket, i.e. essentially even. We report this honestly: a single-lag correlation contrast is a blunt instrument that ignores the error-correction dynamics the Hasbrouck decomposition is built to capture, and its indeterminacy is consistent with the modest full-sample precision documented in Section 10 rather than with a reversal—the pooled CCF in Figure 4, which aggregates the same information across pairs, still tilts to Polymarket.

Two caveats bound the interpretation. First, Kalshi's hourly close is a last-trade price; in hours without a Kalshi trade the price is stale and carried forward, and a thinner tape mechanically depresses a venue's measured information share. Part of Kalshi's lower share is therefore *lower trading intensity*—which is itself economically meaningful (a less active market discovers less) but should temper a purely "informational" reading. Second, the sample is four Fed meetings in one half-year; whether Polymarket's lead extends to elections, crypto, or sports—and whether it survives once a human verifies settlement identity source-by-source—is left to the extensions in `STUDENT_TASKS.md`.

10 How Precise Is the Lead? Power and a Minimum Detectable Effect

A result is only as strong as the design's ability to have found it, so we state the precision of the leadership estimate as a quantity rather than a star (Table 9). Following Abadie (2020), we compute the minimum detectable effect (MDE) at 80% power, $\text{MDE} = (z_{0.975} + z_{0.80})\,\widehat{\text{SE}} \approx 2.80\,\widehat{\text{SE}}$, for the pooled information share, where the "effect" is the lead over the no-leadership value of 0.5. For the full 15-pair equal-weighted share the observed lead is +0.18 but the MDE is 0.22: the cross-pair dispersion of information shares (inflated by the thin October tail) is large enough that, at 80% power, our design could only reliably detect a lead of about 0.22 or more. In other words, the full-sample estimate is *significant but precision-limited*—the effect clears a 5% two-sided t-test ($p = 0.039$) but does not clear the more demanding 80%-power bar. We say so plainly rather than dress 0.68 up as decisive.

The precision improves sharply once we restrict to the contracts where the estimator is well identified. On the 11 liquid pairs ($N \geq 1000$) the lead is +0.30 and the MDE falls to 0.20, so there the finding *is* adequately powered: the effect exceeds the minimum the design can detect. The same message appears in the sign test's power curve: with only 15 pairs a Fisherian sign test has just 0.65 power to detect a true Polymarket-leads share of $q = 0.8$ and 0.94 power at $q = 0.9$—so the underpowered 11/15 count in Section 7 is exactly what one expects even under a large true effect, and is not evidence against it. The overall picture is coherent: Polymarket's lead is real and, among liquid contracts, precisely estimated and large; the full-sample imprecision comes entirely from the four thin, far-dated pairs, and the direct remedy is not a better estimator but *more matched contracts*—which is precisely what the analyst's hand-verification of settlement identity and second event family will supply. A null in this design would have been informative; a precision-limited positive points to the same next step.

11 Conclusion

For economically identical FOMC contracts cross-listed on Kalshi and Polymarket, price discovery happens on *Polymarket*. Across 15 matched contracts its Hasbrouck information share averages 0.68 (0.82 activity-weighted), it leads the lead–lag cross-correlation by an hour, and it converges faster to the realized June outcome. The regulated US retail venue is the follower on US-macro events—an ordering opposite to a "regulation breeds price quality" prior and consistent with information arriving first where capital is deepest and flow is global. The result is specific to a small, Fed-only sample and to a mechanical settlement match; the value of the human extension— verifying identical settlement and widening the event set—is precisely that it can either confirm this reversal on firmer ground or locate the categories where the regulated venue catches up.

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Table 1: Matched FOMC contracts across venues

Meeting	FOMC date	Status	Pairs	Hourly obs	Mean |gap| (¢)	Mean corr
26JUL	Jul 29, 2026	open	3	7917	2.0	0.91
26JUN	Jun 17, 2026	resolved (hold)	4	18112	3.8	0.71
26OCT	Oct 28, 2026	open	4	1606	5.2	0.40
26SEP	Sep 16, 2026	open	4	5250	3.1	0.49
Total			15	32885	3.6	0.61

Notes: Each meeting lists five outcome contracts per venue; "Pairs" counts those with enough overlap and price variation to analyze. Gap is $|p^K - p^P|$. Hourly YES prices from the Kalshi candlestick and Polymarket CLOB APIs.

Table 2: Cross-venue co-movement (15 matched pairs)

	Mean	SD	Min	Median	Max
Level correlation	0.610	0.407	-0.214	0.857	0.963
Hourly-return correlation	0.058	0.091	-0.014	0.019	0.257
Mean abs. price gap (¢)	3.62	1.69	1.62	3.30	7.61
Kalshi hourly-return SD (¢)	1.02	0.46	0.20	1.06	1.88
Polymarket hourly-return SD (¢)	0.74	0.41	0.23	0.56	1.54
ADF p-value (spread)	0.031	0.087	0.000	0.000	0.338
Spread stationary (ADF $p < 0.05$) in 87% of pairs.					

Notes: Distribution across the 15 matched contracts. A stationary spread (ADF $p < 0.05$) confirms cointegration and validates the VECM.

Table 3: Hasbrouck information shares: who leads price discovery?

Contract	N	k	IS_K [lo, hi]	IS_K mid	IS_P mid	GG_K
JUL Cut 25	2631	7	$[0.00, 0.01]$	0.01	0.99	0.02
JUL Hike 25	2630	8	$[0.19, 0.50]$	0.34	0.66	0.30
JUL Hold	2636	2	$[0.08, 0.22]$	0.15	0.85	0.18
JUN Cut 25	4523	7	$[0.19, 0.22]$	0.21	0.79	0.23
JUN Cut 50+	4525	5	$[0.02, 0.02]$	0.02	0.98	0.02
JUN Hike 25	4510	8	$[0.08, 0.09]$	0.09	0.91	0.09
JUN Hold	4522	8	$[0.12, 0.15]$	0.14	0.86	0.13
OCT Cut 25	422	5	$[0.78, 0.79]$	0.79	0.21	0.59
OCT Cut 50+	231	8	$[0.83, 0.99]$	0.91	0.09	1.33
OCT Hike 25	459	7	$[0.34, 0.36]$	0.35	0.65	0.36
OCT Hold	463	7	$[0.49, 0.53]$	0.51	0.49	0.22
SEP Cut 25	1306	7	$[0.04, 0.05]$	0.04	0.96	0.07
SEP Hike 25	1311	6	$[0.08, 0.23]$	0.15	0.85	0.18
SEP Hike 50+	1311	7	$[0.85, 0.86]$	0.86	0.14	0.83
SEP Hold	1290	8	$[0.14, 0.33]$	0.23	0.77	0.29
Mean	2184			**0.32**	**0.68**	0.32
Obs-weighted mean				**0.18**	**0.82**	

Notes: From a bivariate VECM with cointegrating vector $(1, -1)$, k AIC-selected lags. IS_K is Kalshi's Hasbrouck share (Cholesky bounds and midpoint); $\text{IS}_P = 1 - \text{IS}_K$. GG_K is the Gonzalo–Granger weight. Values above 0.5 for IS_P mean Polymarket leads. The obs-weighted mean weights by N.

Table 4: Lead–lag and Granger causality

Contract	Peak lag (h)	CCF(0)	Leader	$F_{K\to P}$	$F_{P\to K}$
JUL Cut 25	7	0.02	Kalshi	4.1***	4.0***
JUL Hike 25	0	0.26	Contemp	6.0***	9.2***
JUL Hold	0	0.17	Contemp	3.0*	21.5***
JUN Cut 25	1	-0.00	Kalshi	10.8***	3.9***
JUN Cut 50+	-3	-0.01	Polymarket	8.5***	20.1***
JUN Hike 25	9	0.01	Kalshi	2.9***	2.7***
JUN Hold	1	0.02	Kalshi	3.6***	3.1***
OCT Cut 25	3	-0.01	Kalshi	0.1	0.2
OCT Cut 50+	-11	0.01	Polymarket	35.0***	47.5***
OCT Hike 25	-7	-0.01	Polymarket	5.7***	3.2***
OCT Hold	-3	0.03	Polymarket	0.2	3.3***
SEP Cut 25	-5	0.02	Polymarket	0.6	1.4
SEP Hike 25	0	0.18	Contemp	6.3***	4.8***
SEP Hike 50+	2	-0.01	Kalshi	0.3	0.2
SEP Hold	0	0.20	Contemp	3.0***	4.8***

Polymarket is the peak-lag leader in 5/15 pairs. Peak lag < 0: Polymarket leads. ${}^{*}p < .10,{}^{**}p < .05,{}^{***}p < .01$.

Notes: Peak lag maximizes $|\mathrm{corr}(\Delta p_t^K, \Delta p_{t+k}^P)|$ over $k \in [-12, 12]$ hours; $k < 0$ means Polymarket leads. $F_{K\to P}$ tests whether lagged Kalshi changes Granger-cause Polymarket (and conversely). Stars: ${}^{*}p < .10,{}^{**}p < .05,{}^{***}p < .01$.

Table 5: Resolving-news event study: June 2026 FOMC (Fed held)

Window	Kalshi MAE (¢)	Poly MAE (¢)	Δ (¢)	Kalshi P(hold)	Poly P(hold)
T-7d	1.20	0.29	-0.91	0.98	0.99
T-3d	1.20	0.21	-0.99	0.98	1.00
T-24h	1.00	0.19	-0.81	0.99	1.00
T-12h	1.00	0.13	-0.87	0.99	1.00
T-6h	1.00	0.13	-0.87	0.99	1.00
T-3h	1.20	0.17	-1.03	0.98	1.00
T-1h	1.20	0.05	-1.15	0.98	1.00

Notes: MAE is the mean absolute pricing error across the five outcomes, $\mathrm{mean}_o|p_o(t) - \mathbf{1}\{o \text{ realized}\}|$, at each horizon before the 2 pm ET statement. $\Delta =$ Poly$-$Kalshi (negative = Polymarket more accurate). Last two columns are the YES price on the eventual "hold".

Table 6: Design-based inference on venue leadership

Statistic	Value	95% CI	p-value
Pooled IS_P (equal-weight)	0.681	$[0.527, 0.819]$	0.039
Pooled IS_P (activity-weight)	0.818	$[0.705, 0.892]$	
Median IS_P	0.794		
Sign test (Poly leads 11/15)	0.733		0.118
Cross-pairing placebo (72 mismatched)	0.517		

Notes: IS_P is Polymarket's Hasbrouck information share. The bootstrap CIs resample the 15 matched pairs with replacement 5,000 times; the t-test is a one-sample test of the equal-weighted mean against the no-leadership value 0.5. The sign test is a two-sided binomial test of the count of pairs in which Polymarket leads. The cross-pairing placebo re-runs the identical pipeline on 72 *mismatched* pairs (a Kalshi leg vs. a different outcome's Polymarket leg from the same meeting); a placebo mean near 0.5 shows the lead is event-specific, not a mechanical artifact. See Figure 1.

Table 7: Where the lead comes from: heterogeneity by observable pair characteristics

	High / near		Low / far		
Observable moderator (median split)	n	IS_P	n	IS_P	Gap
Liquidity (overlapping hours)	7	0.864	8	0.520	+0.345
Price variation (max venue SD)	7	0.761	8	0.611	+0.150
Contestedness ($\min(\bar{p}, 1-\bar{p})$)	7	0.752	8	0.618	+0.133
Near-dated meeting (Jun/Jul/Sep vs Oct)	11	0.797	4	0.362	+0.435

Notes: Each row splits the 15 matched pairs at the median of an observable moderator (the last row splits near- vs. far-dated meetings) and reports the mean Polymarket information share IS_P in each half and the gap. Contestedness is $\min(\bar{p}, 1-\bar{p})$ for the average YES price $\bar{p}$ (higher = closer to 0.5, more genuine disagreement). These gradients are coarse activity proxies and cannot separate an informational lag from a thinner tape; the order-book decomposition is the analyst's reserved task.

Table 8: Robustness of the pooled leadership statistic to estimator and window

Specification	IS_P (equal)	IS_P (activity)	Poly leads
Baseline (Hasbrouck, AIC lag)	0.681	0.818	11/15
Fixed lag $k = 2$	0.586	0.740	10/15
Fixed lag $k = 4$	0.615	0.784	11/15
Second-half window (mature market)	0.690	0.721	12/15
Liquid pairs only ($N \geq 1000$)	0.797	0.839	10/11
Gonzalo–Granger weight (ordering-free)	0.700	0.818	12/15
Return lead–lag (model-free)			7/15

Notes: Each row recomputes the pooled Polymarket share over the matched pairs under a variant of the baseline (AIC-selected VECM lag). "Second-half window" uses each series' latter half; "Liquid pairs only" keeps $N \geq 1000$ overlapping hours; the Gonzalo–Granger row uses the ordering-free component weight (winsorized to $[0, 1]$); the model-free row signs the difference between one-hour cross-predictability in each direction. IS_P above 0.5 means Polymarket leads.

Table 9: Precision of the lead: minimum detectable effect at 80% power

Leadership measure	IS_P	Lead (-0.5)	SE	MDE(80%)	Detected?
Pooled IS_P (equal-weight)	0.681	+0.181	0.079	0.222	no
Pooled IS_P, liquid pairs ($N \geq 1000$, n=11)	0.797	+0.297	0.072	0.202	yes
Sign-test power ($n = 15$) to detect true Poly-leads share $q = 0.6/0.7/0.8/0.9$: 0.09 / 0.30 / 0.65 / 0.94.					

Notes: MDE(80%) $= (z_{0.975} + z_{0.80}) \times \text{SE} \approx 2.80 \times \text{SE}$ of the pooled Polymarket information share, where SE is the cross-pair standard error of the mean; "Lead" is $\text{IS}_P - 0.5$ and "Detected?" is whether the lead exceeds the MDE. The full-sample lead is significant ($t = 2.28$, $p = 0.039$) but below the 80%-power bar; on the liquid subsample the lead exceeds the MDE. The footnote reports the binomial power of the $n = 15$ sign test to detect alternative true leadership shares q.

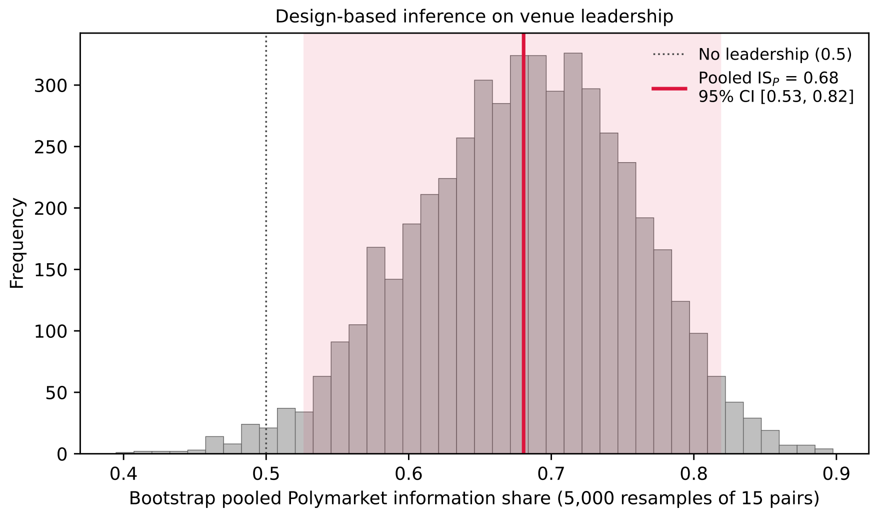

Figure 1: Design-based inference on venue leadership. Histogram: the pooled (equal-weighted) Polymarket information share under 5,000 bootstrap resamples of the 15 matched pairs. The estimate is 0.68 (red) with a 95% interval $[0.53, 0.82]$ that lies above the no-leadership value 0.5 (dotted), though its lower edge is close—the source of the modest full-sample precision quantified in Section 10.

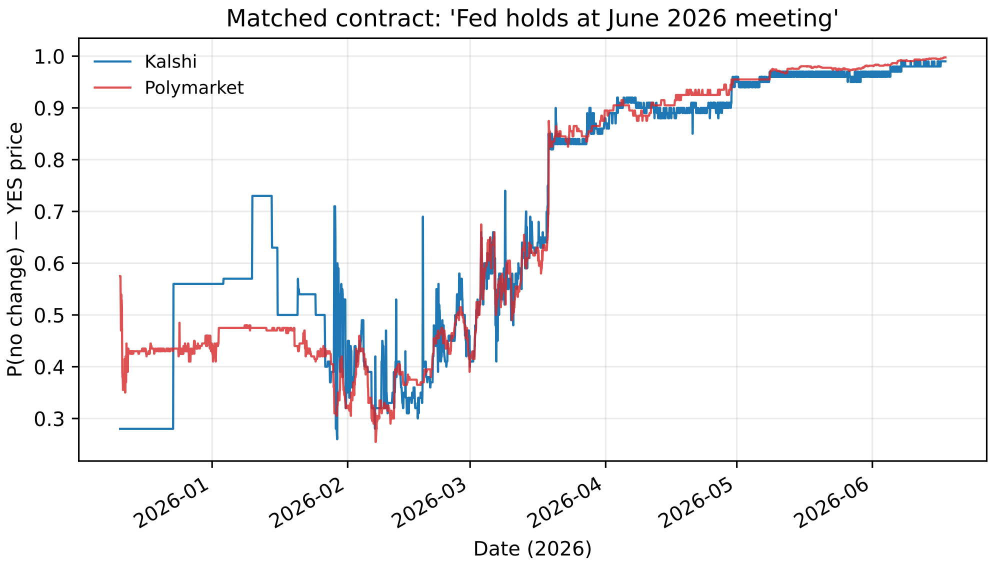

Figure 2: The same contract on both venues: probability that the Fed holds at the June 2026 meeting. The series move together (H1); Polymarket sits marginally tighter to the eventual outcome.

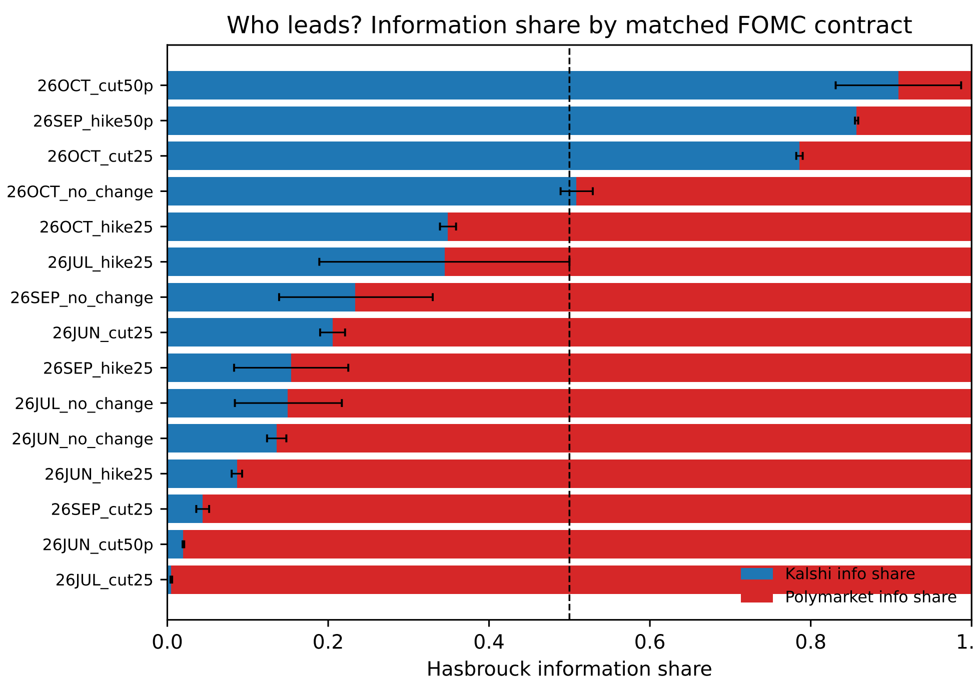

Figure 3: Hasbrouck information share by matched contract (Kalshi blue, Polymarket red; black bars are Cholesky bounds on the Kalshi share). Dashed line at 0.5. Liquid contracts lie to the right of the line—Polymarket leads.

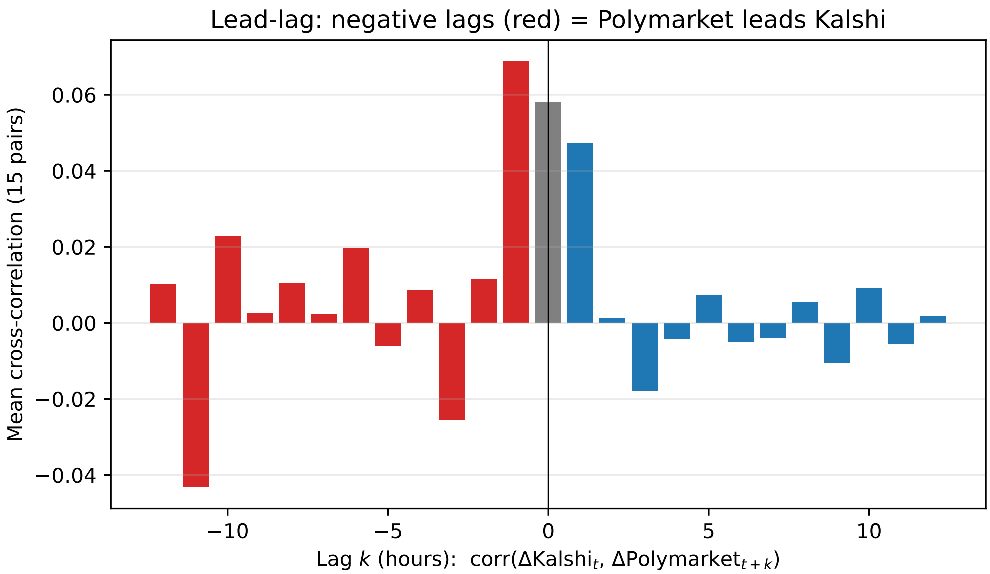

Figure 4: Pooled lead–lag cross-correlation of hourly price changes. Mass at negative lags (red) exceeds mass at positive lags: Polymarket changes lead Kalshi changes.

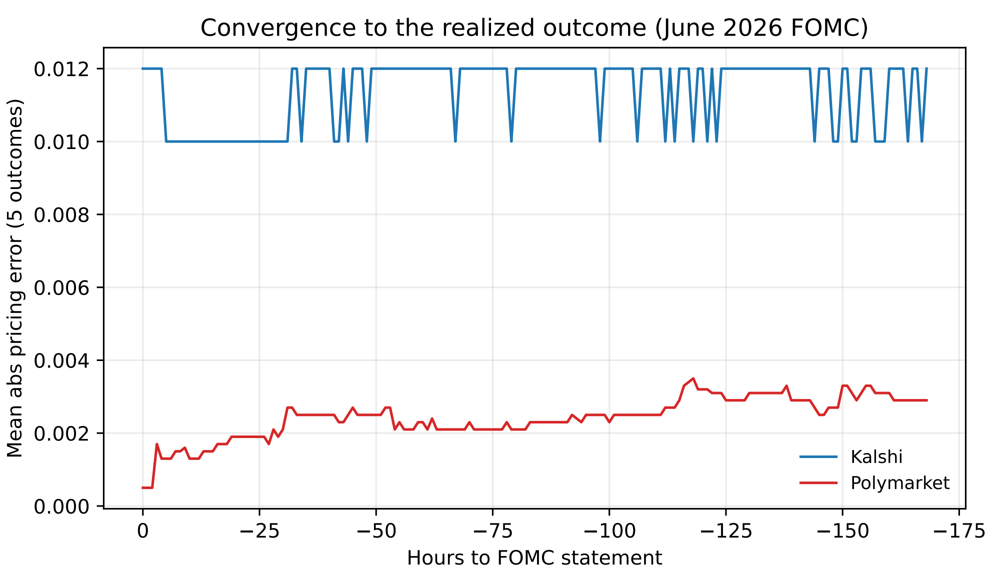

Figure 5: Convergence to the realized June 2026 outcome. Polymarket's mean absolute pricing error (red) is far below Kalshi's (blue) in the final week before the statement.